SOUTHEAST Technical College

Southeast Tech · · Computer Information Systems

Dynamic Website Development CIS-241 30 O

Summer Term 2024-2025 School Year Section 30 O 3.00 Credits 05/19/2025 to 07/25/2025
Modified 05/05/2025

Meeting Times and Location

ONLINE (5/19/2025-7/25/2025)

Contact Information

Instructor: Andy Honey

Email: andrew.honey@southeasttech.edu

Office Hours

Please email me for availability
Online/On Campus TC 209 (by Appointment during the summer)

Course Description

This course addresses the high level of integration of graphics, programming, and dynamic data/information supplied by database management systems (DBMS). The coursework includes discussion and integration of graphics, still images, client and server side scripting languages and data supplied by a DBMS. A PHP/MySQL based electronic portfolio will be developed by each individual student during the course and posted to the Internet to provide potential employers with the student's resume, CIS Programming syllabi and samples of the student's computer programming work in HTML, CSS, JavaScript, PHP, MySQL, C# and Java. Prerequisite: CIS 130, CIS 196

Outcomes

Technical Skills

- WAMP Server
  - Utilize WAMP Server to run Apache and MySQL services.

- Basic PHP Skills
  - Utilize HTML/CSS/JS as appropriate to create a clean UI for end users

    - Utilize PHP include statements to import other files and create page templates
    - Utilize PHP control statements (if/else, try/catch, loops) to structure dynamic code

- PDO Objects and CRUD with MySQL Integration
    - Utilize a PDO Object to connect to a MySQL database with PHP
    - Allow web interface users to perform CRUD operations on the MySQL database

- MVC Pattern/PHP Functions
    - The MVC Pattern is used to create a modular application that is scalable.
    - Code is appropriately placed in Models, Controllers, and Views.

- PHP Sessions/Cookies
    - Sessions are used to maintain and hold data for a user across a session.
    - Cookies are used to hold persistent data across sessions.

- User Accounts/Logins
    - User Accounts are implemented to restrict functionality of the application to only authenticated users.
    - Actions/views are created to allow users to view and modify their information.
    - Login info and some other additional data is stored for users.
    - Passwords are properly stored as hashes, and logins check user passwords.

Professionalism/Communication

- Submission requirements for the assignment are met. This includes video length, file formats, etc.
- Version Control with GitHub
    - Tasks to be completed are created as issues and assigned within the GitHub repository, as well as resolved when completed.
    - Commits show incremental progress, and focus on adding features, fixing bugs, etc.
- Group Work
    - Students will collaborate and assign tasks within a group to complete a single application.

Critical Thinking

- Web Security
    - Applications are analyzed and best practices considered and implemented as appropriate.

Course Materials

Murach's PHP and MySQL

Author: Joel Murach, Ray Harris
Publisher: Murach
Edition: 3rd
ISBN: 978-1-943872-38-1

Deliverables

Homework

Weekly assignments will be used to track attendance. They will be a completion grade, and are intended as student self assessments. The instructor will not make corrections to the homework. It is intended that students study the material provided until they have a good understanding of it.

Unit Project

There are 3 units in this course, each with a Unit Project. Unit Projects will be the assessments for this course, where you will create an application to demonstrate skills, and then present the application to prove your skills. Presentations may be recorded if not attending class on presentation day.

In addition to the presentation, your instructor may require an interview on the Unit Projects to answer follow-up questions and gauge your understanding of the material.

Within Unit Projects, students will be assessed on competencies, which will be used to determine their Project Competencies - Approaching, Proficient, and Mastered scores.

Evaluation Procedures and Grading

Criteria

Please Note: The intent of this grading system is for students to understand each topic in the course in order to pass. Completion of the homework and demonstrated mastery of the course topics is encouraged to raise your grade beyond a C, presuming that you are meeting proficiency in all of the competencies.

Type	Weight	Topic	Notes
Project Competencies - Approaching	40%		Student achieves at least a 2/4 on every project competency
Project Competencies - Proficient	30%		Student achieves at least a 3/4 on every project competency
Project Competencies - Mastery	10%		Each 4/4 score on project competencies, students will receive partial credit towards this score. Example: a student achieving 4/4 on half of the competencies will receive 5% of their final grade weight from this.
Participation/Homework	20%		Homework assignments weighted equally will give partial credit towards this score.

Breakdown

This breakdown is a model/example, not a hard rule for the final grading. Final grading will depend on the grade weighting as defined in the criteria.

Grade	Range	Notes
A	90%-100%	For an A, it is expected that students attend every class session , complete every competency with a score of 3/4 or higher, complete all homework, and completing some competencies with a score of 4/4.
B	80%-89%	For a B, it is expected that students achieve a 3/4 on every competency and also complete most homework assignments, and possibly receive a few 4/4 scores on competencies.
C	70%-79%	C is the minimum passing grade for this course. To receive a C in this course, a student must complete every competency on each assessment with a 3/4 or higher.
D	60% - 69%	Students participating completing all homework but not passing every competency with a 3/4 or higher may expect to receive a D in this course, if they complete every competency with a 2/4.

* Course Expectations

Attendance/Emails

Students will be expected to check emails ***daily.***

Attendance will be measured based on the student's work completion. Students not turning in assignments for 2 consecutive weeks will be marked for administrative withdrawal, per school policy. The instructor will reach out to the student at this time, and if no work is completed by the end of the week, the student will be withdrawn from the course.

Assessment Retakes

Students will be allowed to retake assessments up to one time. Students must confirm the requirements for retakes with their instructor.

No work will be accepted after the Friday that falls on the last week of the course.

Late work for assessments will be counted as your retake for the assessment.

Late Work

No late work will be accepted after the end of the course. Homework/participation assignments will not be accepted late. Unit Projects turned in late will not be allowed retakes.

Collaboration, AI, and Resource Usage

The instructors for CIS courses at Southeast Tech are aware of the existence of external resources such as StackOverflow, ChatGPT, Cursor, etc. However, students are expected to complete their own work on assessments. Students are advised that completing their own homework/activities will help them to prepare for the assessments.

Students will also note that for assessments, instructors may require a meeting to verify their work or answer additional follow-up questions to demonstrate knowledge of competencies. Because of this, students should understand what their work does and why it does it, so that they can answer emergent questions pertaining to their work.

Cheating/Plagirism

Students are expected to complete their own work on assessments, using references as models for how to complete work, and not for copying/taking directly from resources.

Instructors reserve the right to issue a zero for work suspected of cheating. Cheating includes, but is not limited to:

- Utilizing AI to write large portions of code
- Copying large sections of code from external sources
- Plagiarizing the work of others, or copying work from another student

Student use of AI and other online resources should be limited to research and filling in the gaps in their knowledge of the material. For example, asking an AI model for ideas on how to structure an application and then independently recreating those ideas from your own understanding is acceptable use of AI.

Professionalism

Projects in this course may include a degree of freedom in choosing the content/topics presented within them. In preparing for a career, professionalism with content choice is an important area of focus.

Projects including unprofessional topics will be docked 20-100% depending on the severity, and at the instructor's discretion. Intent and context will also be considered.

If a student's work is to be presented in class, the instructor reserves the right to suspend a presentation for the content shown.

Content that should be considered unprofessional includes, but is not limited to:

- Violence/gore/viscera
- Illicit substances
- Non-prescription drugs
- Alcohol
- Non-prescription stimulants
- Content of a pornographic nature

- Content that harasses or discriminates against a person or group of people
- Slurs or slang commonly considered inappropriate
- Content that may be seen as vulgar/offensive/obscene by some audiences

In summary, please try to keep content appropriate for a PG rating. We are all (hopefully) adults, but we are also professionals. Professionalism doesn't have to be boring, but it does have to be mature in the way that it handles content.

If there are any questions about what should/should not be included in your assignments, please contact the instructor for this course, as they would be happy to work with you on what is acceptable.

Course Outline

When	Topic	Notes
Unit 1 Weeks 1-3	Intro to PHP/MySQL	• WAMP Server • Basic PHP • Basic MySQL • PDO Objects
Unit 2 Weeks 4-7	Dynamic Web Applications	• The MVC Format • Web Forms • Arrays and Control Statements • Cookies and Sessions • Functions and Libraries
Unit 3 Weeks 8-10	Web Security	• Objects in PHP • Regular Expressions/Data Validation • Exception Handling • Reading/Writing Files • Secure Socket/Encryption • Login Systems

Institutional Policies

Course Subject to Change

The syllabus is a statement of intent and schedule of topics, activities, assessments, and requirements as well as an overall road map to ensure students will achieve both the course learning objectives and the program-level learning objectives.

The scheduled educational activities are subject to change with reasonable advance notice to students by either the instructor or academic administration. The grading and attendance policies are not subject to change once posted.

During campus closures, the dynamic nature of our learning environment supports the possibility of remote learning days. In such instances, instructors will demonstrate flexibility, providing necessary support to ensure students can actively engage in coursework while accommodating individual circumstances. Clear communication and understanding will be prioritized to maintain the quality of education.

Students wishing to appeal changes to a course syllabus must first do so in writing to their instructor. If a successful resolution is not achieved after submitting the written appeal, the student has the option to file a written appeal to the appropriate Associate Dean of Curriculum and Instruction or Dean of Curriculum and Instruction. If the second level appeal does not result in a resolution, the student can submit a final appeal to the Vice President of Academic Affairs. The review and final decision of the Vice President of Academic Affairs is final and not subject to appeal.

Attendance

Administrative Withdraw for Non-Attendance

Students are expected to attend all sessions of courses for which they are enrolled. Absences do not excuse the student from meeting course requirements. Each instructor will evaluate the student's progress to determine how to handle outstanding assignments and assessments. Students attending 16 week courses who have a record of zero attendance for fourteen (14) calendar days will be administratively withdrawn from the course(s) the absence is occurring, resulting in a grade of "W" for the course. Students attending courses of less than a 16 week duration, who have a record of zero attendance for seven (7) calendar days will be administratively withdrawn. Students who receive financial aid may have their financial aid status affected. More information is located in the Southeast Tech Catalog (https://catalog.southeasttech.edu/index.php):

Course Recording Statement

Delivery of some courses at STC allows flexible participation options for students in time and location. Courses may be made available to properly enrolled students who are not physically in the classroom, and/or a time delayed recording made available to all properly enrolled students. In addition to instructional content, remote delivery may include capture of all classroom audio and visual happenings during designated class times, including that of students who may be part of the class, but who are not physically in the classroom. By remaining in classes capturing audio and visual happenings, you are agreeing to this statement.

Institutional Learning Outcomes

Student success is important to Southeast Tech faculty, and all faculty are involved in assessing student learning. Upon graduation, Southeast students will have competence in the following four institutional learning outcomes:

Technical Skills: Students will be able to explain industry-relevant concepts (knowledge) and demonstrate industry-relevant technical skills (performance).

Communication: Students will be able to define the purpose of the communication; organize and structure the communication; provide supporting material; demonstrate precision of language; and professionally deliver and format the communication.

Problem Solving & Critical Thinking: Students will be able to define the problem; analyze the problem; generate solutions; evaluate solutions; and select the best solution.

Professionalism: Students will be able to demonstrate positive work ethic; collaborate as part of a team; adapt to change; adhere to professional standards; and model integrity and ethics.

Student Conduct, Cheating and Plagiarism

Students attending Southeast Technical College are expected to follow policies and requirements in the Southeast Technical College Student Handbook: STC Student Handbook (https://catalog.southeasttech.edu/index.php). In addition, some departments have academic handbooks for their areas that outline expected student conduct.

Southeast policies authorize the Vice President of Academic Affairs to suspend or terminate any student from the college for misconduct as outlined in the rules of this policy. The following student misconduct shall constitute grounds for student discipline, suspension, or termination when such activity occurs on school grounds, in the online course environment or during an educational function under the auspices of the school board: acts of dishonesty, including cheating and plagiarism or other forms of dishonesty relating to academic achievement such as the misuse of technology, software, apps, or other computer-assisted or artificial intelligence in violation of individual course or assignment policies as defined by instructors.

Test Proctoring

The Southeast Testing Center offers test proctoring located in the Wood Center Building, Room 103. Online instructors will communicate details regarding required test proctoring and how to sign up to take these assessments. Testing Center contact information: 605.367.6014 or testing@southeasttech.edu

Academic Resource Center (ARC)

The Academic Resource Center (ARC) is dedicated to helping Southeast Tech students achieve their academic goals and pursue their professional ambitions. While classes and assignments can sometimes pose challenges, the ARC provides essential support through free one-on-one and online tutoring from both professional and peer tutors. Services include academic assistance with course content, opportunities to practice learned material, strategies for independent learning, personalized feedback, and study skills development in areas such as writing, math, test-taking, memorization, note-taking, and organization. Additionally, the ARC offers techniques to reduce anxiety and stress related to assignments, methods to boost academic confidence and self-esteem, and distraction-free workspaces to enhance focus. Learn more about these services by visiting the ARC page found in the Student tab of

myTech: Academic Resource Center and Resources on myTech (https://my.southeasttech.edu/ICS/Student/Student_Support__Services/Student_Success_Center/Academic_Resource_Center/)

Student Mental Health

Mental health and overall well-being are significant predictors of academic success. As such, it is essential that during your college experience you develop the skills and optimize resources effectively to navigate stress, anxiety, depression, and other mental health concerns. Please familiarize yourself with the range of resources Southeast Tech provides and seek out support for mental health concerns as needed. Individual counseling services are available in the Sullivan Health Science Center or virtually via Teams-based video, Telephone, or Email. Please visit Personal Counseling on myTech (https://my.southeasttech.edu/ICS/Student/Student_Support__Services/Student_Success_Center/Personal_Counseling.jnz) to complete the Personal Counseling form. Contact the counselor at counseling@southeasttech.edu (mailto:counseling@southeasttech.edu) or 605-367-4821 for more information and to schedule a session. Crisis Counseling is available nationwide by calling 988, and in the Sioux Falls area via the Helpline Center at Helpline Center Website (https://www.helplinecenter.org/)

Safety

Violation of safety to self and others and/or violations of safe operating practices of equipment may result in the reduction or loss of a daily grade, removal from class, and/or other disciplinary action.

ADA and Accessibility Services

It is the policy and practice of Southeast Technical College and the instructor of this course to act with integrity, engage in equitable conduct with respect to differences in age, gender, race, disability, and religion, and to create inclusive and accessible learning environments consistent with federal and state laws. In accordance with Section 504 of the Rehabilitation Act of 1973, the Americans with Disabilities Act of 1990, and the Americans with Disabilities Amendment Act of 2008, Southeast Technical College ensures that no student with a disability will be discriminated against or denied access to participate in, or benefit from, programs, services and activities. If you are a student with a documented disability, think you may have a disability, or have a temporary health condition that requires accommodations, please visit the Accessibility Services (https://www.southeasttech.edu/student-life/accessibility-services.php) website, call 605-367-6110, or email access@southeasttech.edu (mailto:access@southeasttech.edu). All students are encouraged to discuss access-related needs with their instructors and the Accessibility Services Office as soon as possible in their academic career.

Freedom of Speech

Southeast Tech strives to create an environment in which diverse opinions can be expressed and heard. Students have the right to peacefully express their views and opinions, regardless of whether others may disagree, but not in such a way as to interfere with the rights of others or the operation of the college. Views expressed should not violate any of Southeast Tech policies or core values.

Southeast Technical College students are both citizens and members of the Southeast Tech community. Students should enjoy the same freedom of speech, peaceful assembly, and right to petition that other citizens enjoy and should abide by the obligations and expectations as members of the Southeast Tech community.

Notice of Nondiscriminatory Policy

Southeast Tech is committed to a policy of equal opportunity for all in every aspect of its operations. Southeast Tech does not discriminate on the basis of race, color, creed, religion, age, gender (including pregnancy), sexual orientation, disability, national origin or ancestry, military/veteran status, genetic information, or any other category protected by law in its educational programs, admissions policies, employment policies, financial aid, or other institute-administered programs. Click this link for more information: Non-Discriminatory Policy and Contact Information (https://catalog.southeasttech.edu/index.php)

Title IX/Pregnancy Leave/Sexual Misconduct/Sexual Assault/Sexual Offense

Southeast Tech is committed to providing a learning and working environment free of unlawful harassment based on an individual's race, color, religion, creed, ancestry, national origin, gender, sexual orientation, disability, age, protected military/veteran status, genetic information or any other basis protected by law. Faculty also work with students to provide medically necessary student absences related to pregnancy, childbirth, related conditions, and recovery. Concerns may be reported directly to this individual in person, by calling 605-367-5681, or by emailing Jim.Larson@southeasttech.edu. Individuals with immediate concerns should call campus security at 605-367-7847. For more information, click this link: Title IX in Student Catalog (https://catalog.southeasttech.edu/index.php)

Sexual Misconduct

Southeast Tech is committed to providing a learning and working environment that emphasizes the dignity and worth of every member of its community, free from discriminatory conduct. Sexual harassment in any form or context is contrary to this commitment and will not be tolerated. Sexual harassment subverts the mission and the work of Southeast Tech and can threaten the career, educational experience, and well-being of students, faculty, and staff.

Southeast Tech recognizes that sexual harassment includes harassment based on gender, sexual orientation, gender identity, or gender expression and may occur regardless of the formal position or status of each person involved. Sexual harassment is especially offensive, however, when it occurs in relationship between an instructor and student or between supervisor and subordinate. In those situations, sexual harassment exploits unfairly the power inherent in the position of the faculty member or supervisor.

Sexual harassment also constitutes a form of sexual discrimination that is illegal under Title VII of the Civil Rights Act of 1964, Title IX of the Elementary/Secondary Education Act of 1972, as amended, and state law. Southeast Tech also recognizes that sexual assault, domestic violence, dating violence, and stalking may also be considered sexual harassment.

Students have the right to be free of sexual misconduct while pursuing their education at Southeast Tech which would include participating in any off-campus activity such as internships, clinical experiences, and preceptor experiences.

Students should immediately report any form of sexual misconduct to their instructor or the designated contact at Southeast Tech.

Student Right-to-Know/Campus Security Act/Clery Act

Southeast Tech is required under Public Law 101-542 to encourage students to report all crimes (murder, rape, forcible or non-forcible sex offenses, robbery, aggravated assault, burglary, motor vehicle and other thefts or crimes) which occur on the Southeast Tech Campus to the local law enforcement offices and to the Southeast Tech Student Success Center or campus security. For more information on Student Right to Know/Campus Security Act, click this link: Student Right to Know - Campus Security Act (https://catalog.southeasttech.edu/index.php)

Family Educational Rights & Privacy Act

FERPA stands for the Family Educational Rights and Privacy Act of 1974. It is a federal law governing the privacy and handling of educational records and giving specific rights to students. You can find more information about the law at FERPA Website (https://studentprivacy.ed.gov/ferpa)

Student Communications

Southeast Tech's student email accounts will be used as Southeast Tech's official means of communications with students. Southeast Tech students should also use the myTech Student and Campus Life tabs for information regarding important upcoming events and required dates for fee payment, etc. The Student and Campus Life tabs will also inform students about job opportunities, scholarship information, and other important announcements.